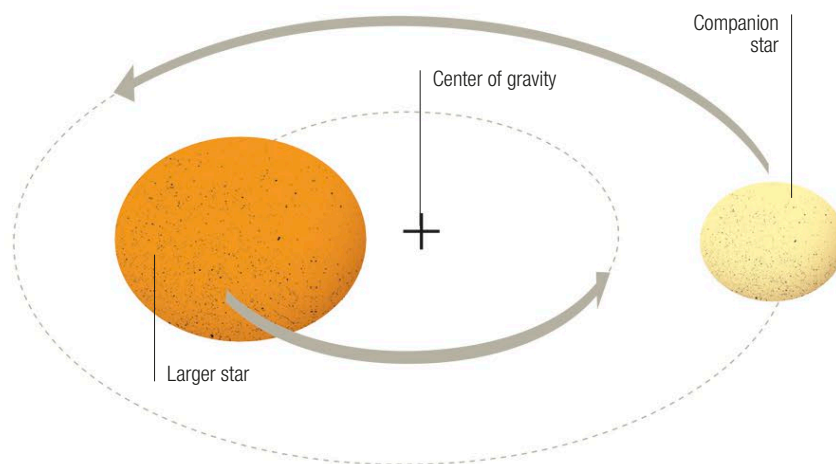


Binary star systems

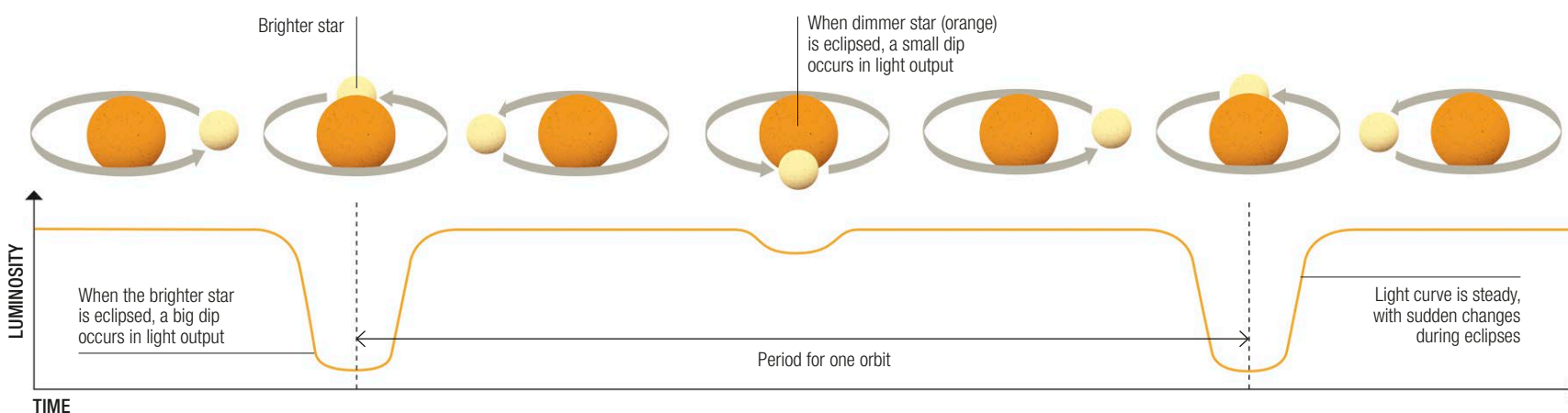
Extrinsic variable stars owe their apparent variations in brightness to something other than changes in light output. The most important group of extrinsic variables are eclipsing binaries. These are binary systems (two stars orbiting each other) with an orbital plane that lines up with Earth. From time to time, one star eclipses (blocks out light from) the other as seen from Earth, causing some dimming. A slight dimming occurs when the brighter star eclipses the fainter star, and a more significant dimming when the fainter star eclipses the brighter one. The first eclipsing binary to be discovered was Algol, in the constellation Perseus. This actually consists of three stars, of which two regularly eclipse each other. Each time the fainter of the two eclipses the brighter, which occurs every 2.86 days, there is a roughly 70 percent dimming for about 10 hours.

A different and somewhat unusual cause of extrinsic variability occurs where two closely orbiting stars in a binary system have acquired distorted, ellipsoidal shapes. These are called rotating ellipsoidal binaries (see right). An example is the bright star Spica (actually a pair of stars) in the constellation Virgo.



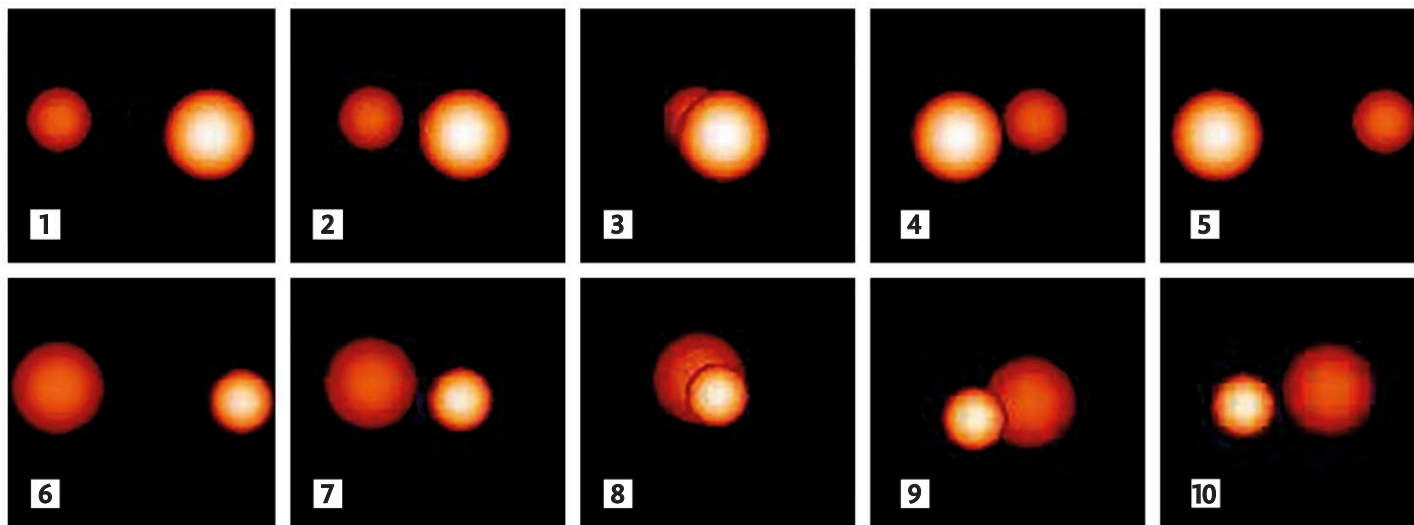
△ Eclipsing and ellipsoidal variables

In this type of variable, two stars that are orbiting a common center of gravity become distorted into ellipsoidal (egglike) shapes. Sometimes they appear side-on (as here) and at other times end-on (appearing smaller and rounder), which affects how bright they look from Earth.



△ Light curve of an eclipsing binary

Eclipsing binary stars are detected by regularly occurring apparent dips in a star's brightness. These dips in brilliance occur when one of a pair of stars partially blocks the light coming from the other star, as seen from Earth. The biggest dip occurs when the dimmer of the two stars eclipses the brighter star.



◁ Binary orbit sequence

These 10 frames from a movie made with a special infrared-sensitive camera show two young stars orbiting a shared center of gravity. The images were taken using the ADaptive Optics Near Infrared System (ADONIS) at the European Southern Observatory at La Silla, Chile.